

push protocol. The puzzle is completed by introducing a special mail access protocol that transfers messages from Bob's mail server to his local PC. There are currently a number of popular mail access protocols, including **Post Office Protocol—Version 3 (POP3)**, **Internet Mail Access Protocol (IMAP)**, and HTTP.

Figure 2.18 provides a summary of the protocols that are used for Internet mail: SMTP is used to transfer mail from the sender's mail server to the recipient's mail server; SMTP is also used to transfer mail from the sender's user agent to the sender's mail server. A mail access protocol, such as POP3, is used to transfer mail from the recipient's mail server to the recipient's user agent.

POP3

POP3 is an extremely simple mail access protocol. It is defined in [RFC 1939], which is short and quite readable. Because the protocol is so simple, its functionality is rather limited. POP3 begins when the user agent (the client) opens a TCP connection to the mail server (the server) on port 110. With the TCP connection established, POP3 progresses through three phases: authorization, transaction, and update. During the first phase, authorization, the user agent sends a username and a password (in the clear) to authenticate the user. During the second phase, transaction, the user agent retrieves messages; also during this phase, the user agent can mark messages for deletion, remove deletion marks, and obtain mail statistics. The third phase, update, occurs after the client has issued the `quit` command, ending the POP3 session; at this time, the mail server deletes the messages that were marked for deletion.

In a POP3 transaction, the user agent issues commands, and the server responds to each command with a reply. There are two possible responses: `+OK` (sometimes followed by server-to-client data), used by the server to indicate that the previous command was fine; and `-ERR`, used by the server to indicate that something was wrong with the previous command.

The authorization phase has two principal commands: `user <username>` and `pass <password>`. To illustrate these two commands, we suggest that you Telnet directly into a POP3 server, using port 110, and issue these commands. Suppose that `mailServer` is the name of your mail server. You will see something like:

```
telnet mailServer 110
+OK POP3 server ready
user bob
+OK
pass hungry
+OK user successfully logged on
```

If you misspell a command, the POP3 server will reply with an `-ERR` message.

Now let's take a look at the transaction phase. A user agent using POP3 can often be configured (by the user) to “download and delete” or to “download and keep.” The sequence of commands issued by a POP3 user agent depends on which of these two modes the user agent is operating in. In the download-and-delete mode, the user agent will issue the `list`, `retr`, and `dele` commands. As an example, suppose the user has two messages in his or her mailbox. In the dialogue below, `C:` (standing for client) is the user agent and `S:` (standing for server) is the mail server. The transaction will look something like:

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: (blah blah ...
S: .....
S: .....blah)
S: .
C: dele 1
C: retr 2
S: (blah blah ...
S: .....
S: .....blah)
S: .
C: dele 2
C: quit
S: +OK POP3 server signing off
```

The user agent first asks the mail server to list the size of each of the stored messages. The user agent then retrieves and deletes each message from the server. Note that after the authorization phase, the user agent employed only four commands: `list`, `retr`, `dele`, and `quit`. The syntax for these commands is defined in RFC 1939. After processing the `quit` command, the POP3 server enters the update phase and removes messages 1 and 2 from the mailbox.

A problem with this download-and-delete mode is that the recipient, Bob, may be nomadic and may want to access his mail messages from multiple machines, for example, his office PC, his home PC, and his portable computer. The download-and-delete mode partitions Bob's mail messages over these three machines; in particular, if Bob first reads a message on his office PC, he will not be able to reread the message from his portable at home later in the evening. In the download-and-keep mode, the user agent leaves the messages on the mail server after downloading them. In this case, Bob can reread messages from different machines; he can access a message from work and access it again later in the week from home.

During a POP3 session between a user agent and the mail server, the POP3 server maintains some state information; in particular, it keeps track of which user messages have been marked deleted. However, the POP3 server does not carry state information across POP3 sessions. This lack of state information across sessions greatly simplifies the implementation of a POP3 server.

IMAP

With POP3 access, once Bob has downloaded his messages to the local machine, he can create mail folders and move the downloaded messages into the folders. Bob can then delete messages, move messages across folders, and search for messages (by sender name or subject). But this paradigm—namely, folders and messages in the local machine—poses a problem for the nomadic user, who would prefer to maintain a folder hierarchy on a remote server that can be accessed from any computer. This is not possible with POP3—the POP3 protocol does not provide any means for a user to create remote folders and assign messages to folders.

To solve this and other problems, the IMAP protocol, defined in [RFC 3501], was invented. Like POP3, IMAP is a mail access protocol. It has many more features than POP3, but it is also significantly more complex. (And thus the client and server side implementations are significantly more complex.)

An IMAP server will associate each message with a folder; when a message first arrives at the server, it is associated with the recipient's INBOX folder. The recipient can then move the message into a new, user-created folder, read the message, delete the message, and so on. The IMAP protocol provides commands to allow users to create folders and move messages from one folder to another. IMAP also provides commands that allow users to search remote folders for messages matching specific criteria. Note that, unlike POP3, an IMAP server maintains user state information across IMAP sessions—for example, the names of the folders and which messages are associated with which folders.

Another important feature of IMAP is that it has commands that permit a user agent to obtain components of messages. For example, a user agent can obtain just the message header of a message or just one part of a multipart MIME message. This feature is useful when there is a low-bandwidth connection (for example, a slow-speed modem link) between the user agent and its mail server. With a low-bandwidth connection, the user may not want to download all of the messages in its mailbox, particularly avoiding long messages that might contain, for example, an audio or video clip.

Web-Based E-Mail

More and more users today are sending and accessing their e-mail through their Web browsers. Hotmail introduced Web-based access in the mid 1990s. Now Web-based